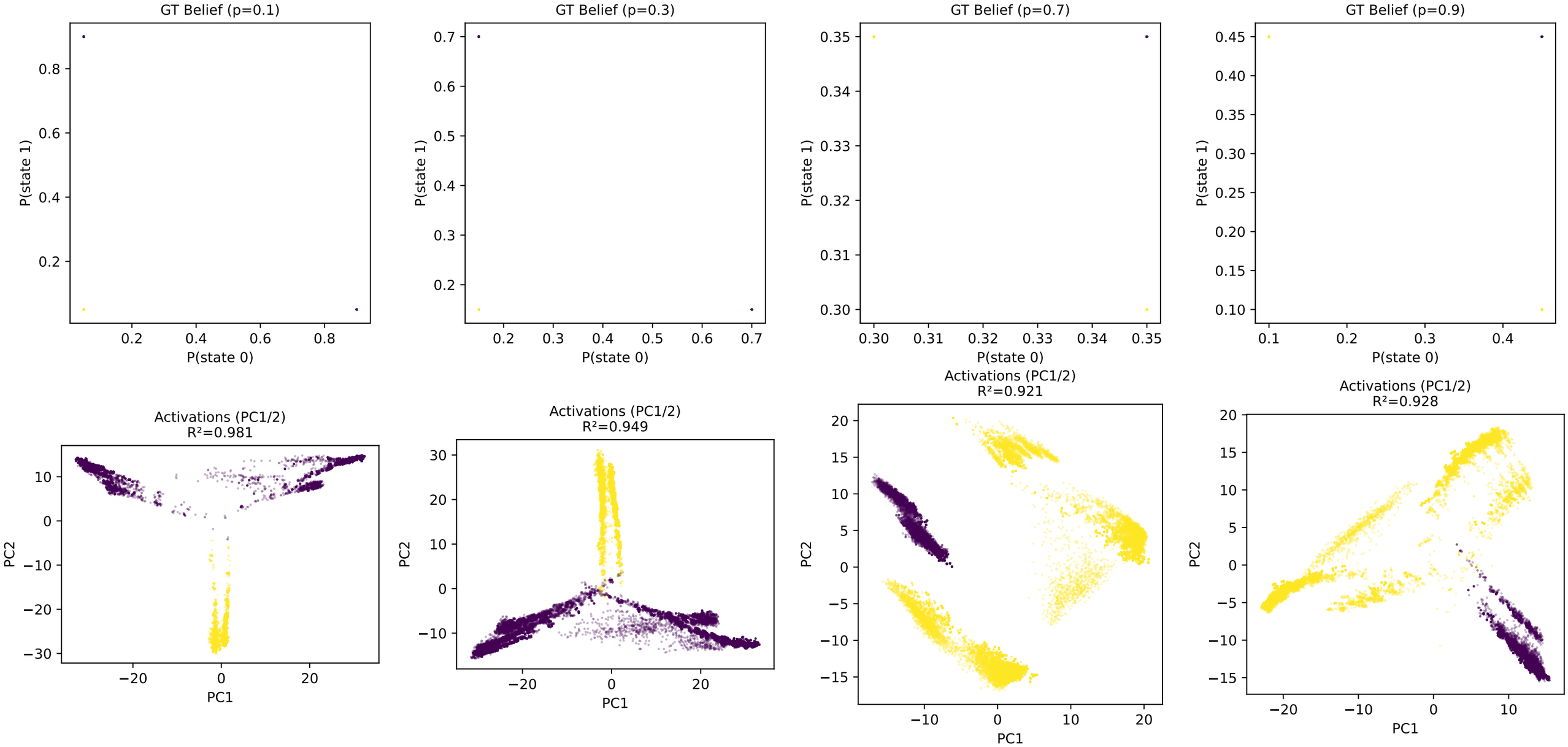K4: Belief Geometry Recovery